CARTIER DIVISOR CASE: MULTIPLICITY EQUALS THE LENGTH CUT OUT BY THE EQUATION

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0B04: sources/stacks-chow/tags.txt, source lines 13847–13847. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02MC: sources/stacks-chow/tags.txt, source lines 3439–3439. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Lemma 0B04 assume that $c = 1$, i.e., V is an effective Cartier divisor. Then*

$$e(X, V \cdot W, Z) = length_{\mathcal{O}_{X,\xi}}(\mathcal{O}_{W,\xi}/f_1\mathcal{O}_{W,\xi}).$$

Proof. In this case the image of f_1 in $\mathcal{O}_{W,\xi}$ is nonzero by properness of intersection, hence a nonzerodivisor divisor. Moreover, $\mathcal{O}_{W,\xi}$ is a Noetherian local domain of dimension 1. Thus

$$\text{length}_{\mathcal{O}_{X,\xi}}(\mathcal{O}_{W,\xi}/f_1^t\mathcal{O}_{W,\xi}) = t\text{length}_{\mathcal{O}_{X,\xi}}(\mathcal{O}_{W,\xi}/f_1\mathcal{O}_{W,\xi})$$

for all $t \geq 1$, see Algebra, Lemma 02MC. This proves the lemma. □

Date: stacks-project:intersection.tex:local-snapshot-20261012; GNU FDL 1.2 or later.